

Image Enhancement



- Fourier Transform
- Convolution based filtering
- Non-linear filtering
- Image sharpening in frequency domain.



Fourier Transform

Fourier Transforms



Fourier transform is an extremely powerful mathematical tool that allows to view the signal in the different domain (Frequency)

This allows to easily pick and choose the most important components and remove all the noise.

Image processing

The Fourier transform gives information about what is happening to image in frequencies terms.

- Certain operation are easier to perform in frequency domain
- Image transform helps to design faster algorithms
- Eliminating high frequencies give image blurs
- Eliminating low frequencies gives image edges
- Enhancing high frequencies and keeping the low frequencies leads to image sharpness

Applications in computer vision



Image Filtering:

By manipulating the frequencies in the frequency domain (e.g., removing high frequencies to blur an image or enhancing high frequencies to sharpen it), you can achieve various filtering effects.

Noise Reduction:

High-frequency noise can be **identified and suppressed** in the frequency domain.

Image Compression:

By discarding the **less significant high-frequency components**, the image can be compressed.

Blur Detection:

A blurry image will have a lower concentration of high-frequency components compared to a sharp image, which can be detected by **analyzing the magnitude of the FFT**.

Feature Extraction:

The Fourier transform can **highlight periodic patterns and texture**, which can be used as **features for tasks like shape recognition**.

Jean Baptiste Josef Fourier



(1768-1830)



$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

where,

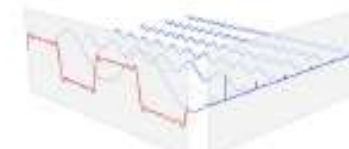
- $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$
- $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$
- $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$
- $n = 1, 2, 3, \dots$

French mathematician, known also as an Egyptologist and administrator, who exerted strong influence on mathematical physics

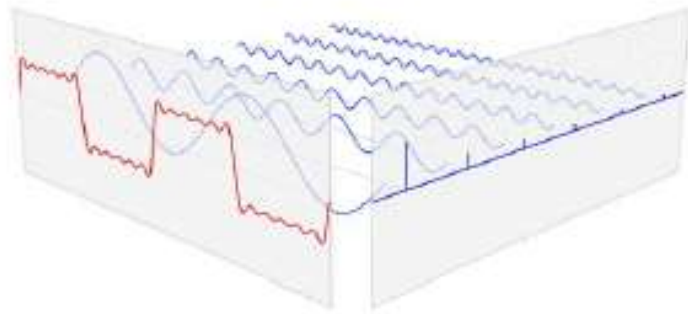
He showed how the conduction of heat in solid bodies may be analyzed in terms of infinite mathematical series now called by his name, the Fourier series

Any Periodic Function can be rewritten as a Weighted Sum of Infinite Sinusoids of different frequencies.

Egyptologist: study all the aspects of ancient Egypt that they can, across periods from about 7.000 BC to the early middle aoes.



Basic building block:



infinite sum of sine waves

$$A \sin(\omega x + \phi)$$

Diagram illustrating the components of the sine wave equation $A \sin(\omega x + \phi)$:

- A : amplitude
- $\sin$: sinusoid
- ω : angular frequency
- x : variable
- ϕ : phase

Fourier's claim: Add enough of these to get any periodic signal you want!



Fourier transform

inverse Fourier transform

continuous

$$F(k) = \int_{-\infty}^{\infty} f(x) e^{-j2\pi kx} dx$$

$$f(x) = \int_{-\infty}^{\infty} F(k) e^{j2\pi kx} dk$$

discrete

$$F(k) = \frac{1}{N} \sum_{x=0}^{N-1} f(x) e^{-j2\pi kx/N}$$

$k = 0, 1, 2, \dots, N-1$

$$f(x) = \sum_{k=0}^{N-1} F(k) e^{j2\pi kx/N}$$

$x = 0, 1, 2, \dots, N-1$

Where is the connection to the ‘summation of sine waves’ idea?

Fourier transform



Where is the connection to the ‘summation of sine waves’ idea?

$$f(x) = \sum_{k=0}^{N-1} F(k) e^{j2\pi kx/N}$$

Euler's formula

$$e^{j\theta} = \cos \theta + j \sin \theta$$

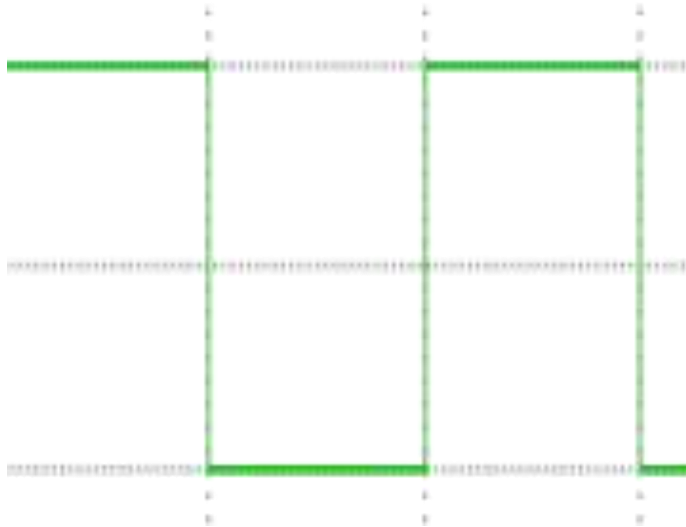
sum over frequencies

$$f(x) = \sum_{k=0}^{N-1} F(k) \left\{ \cos(2\pi kx) + j \sin(2\pi kx) \right\}$$

scaling parameter

wave components

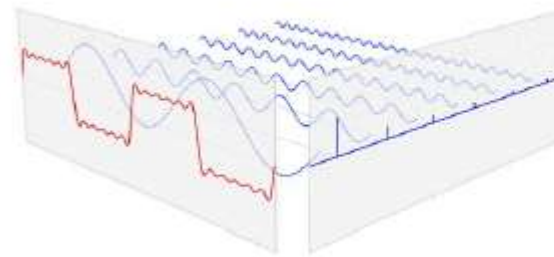
Square wave equation:



square wave

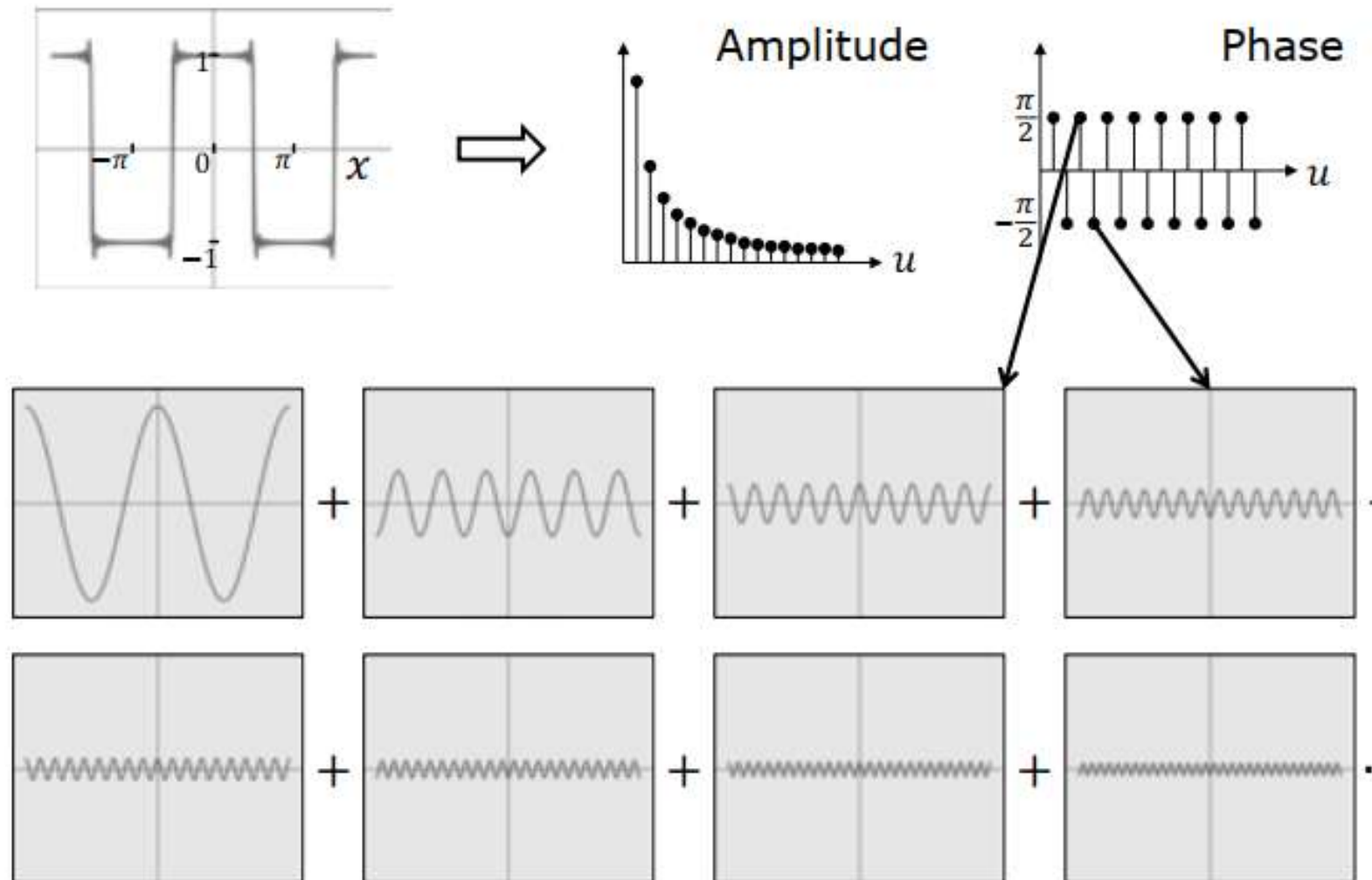
=

$$A \sum_{k=1}^{\infty} \frac{1}{k} \sin(2\pi kx)$$



infinite sum of sine waves

Frequency spectrum of Square wave:

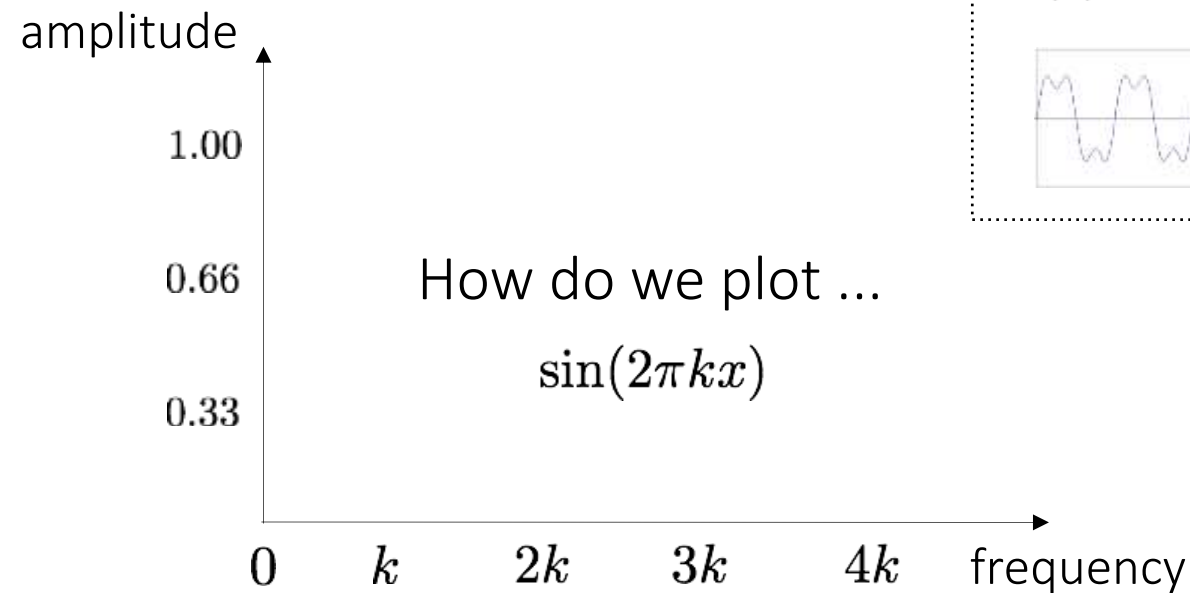




Frequency domain

Visualizing Frequency Spectrum:

Recall the temporal domain visualizati

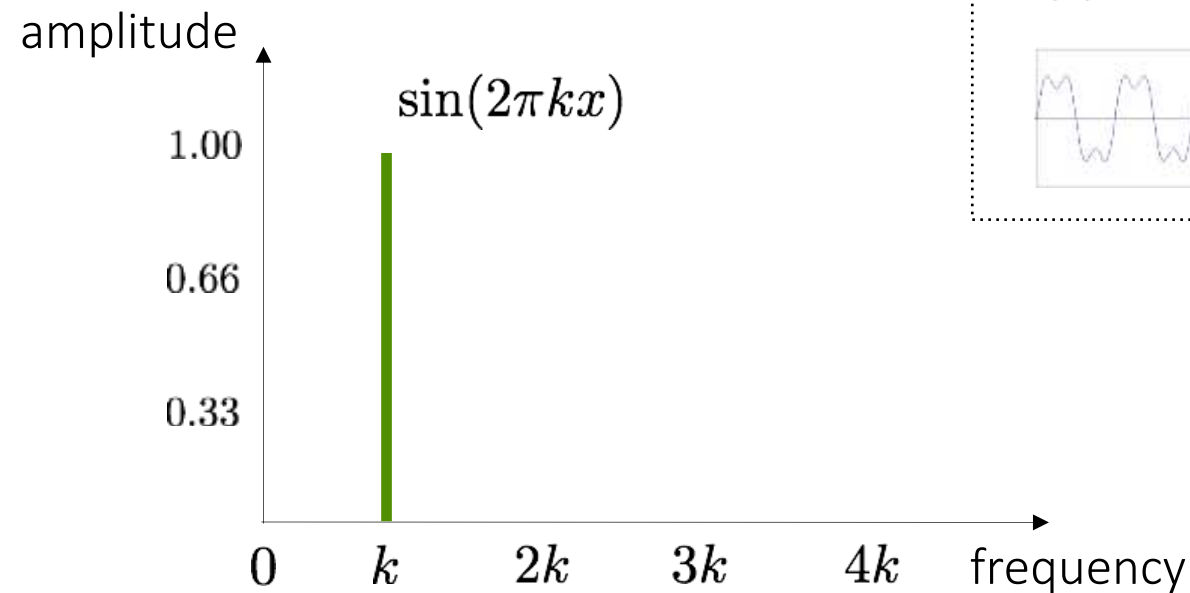


$$f(x) = \sin(2\pi kx) + \frac{1}{3} \sin(2\pi 3kx)$$



Visualizing Frequency Spectrum:

Recall the temporal domain visualizati

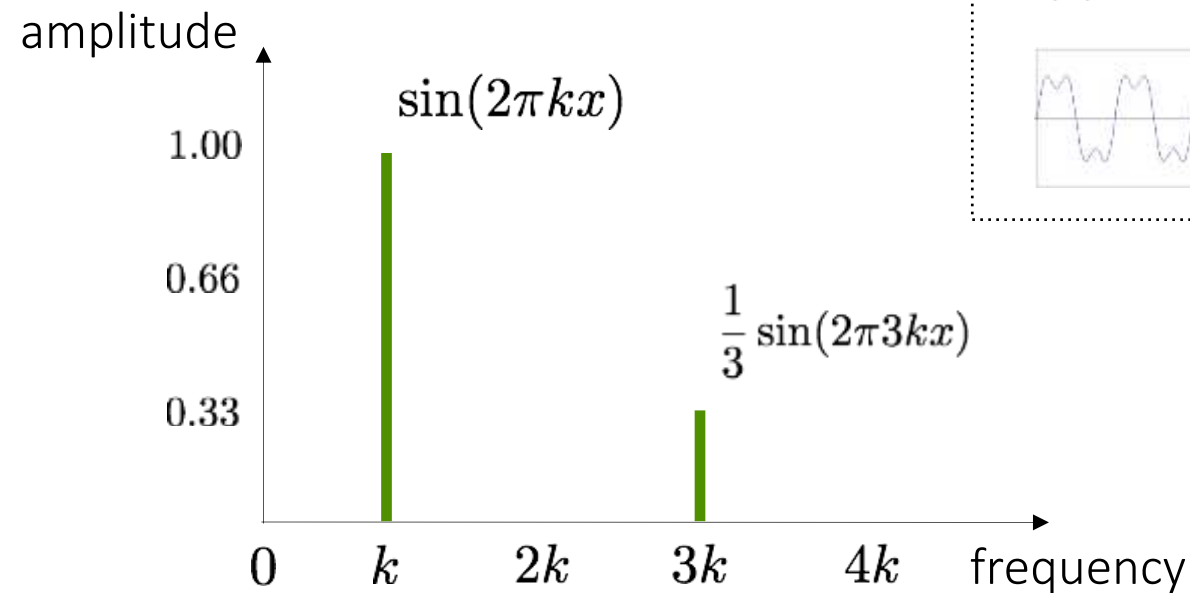


$$f(x) = \sin(2\pi kx) + \frac{1}{3} \sin(2\pi 3kx)$$



Visualizing Frequency Spectrum:

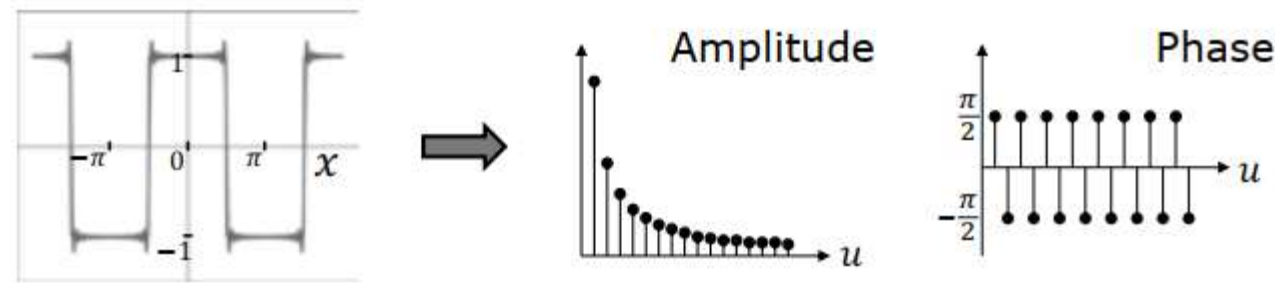
Recall the temporal domain visualizati



$$f(x) = \sin(2\pi kx) + \frac{1}{3} \sin(2\pi 3kx)$$



Fourier Transform (FT):



The magnitude represents the strength of each frequency component

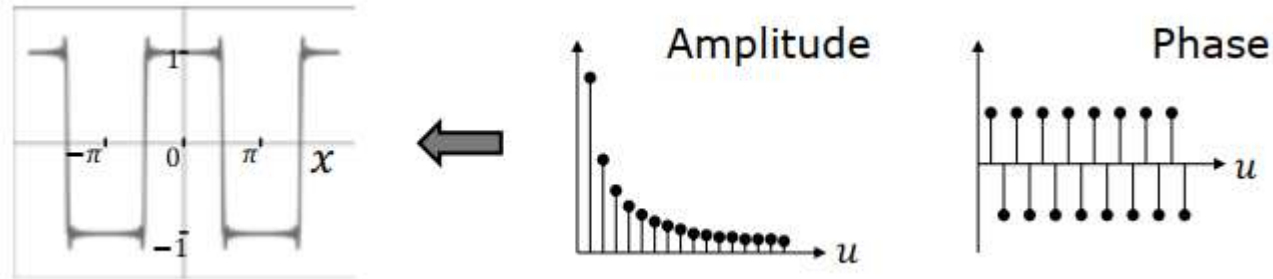
phase represents its position or location and orientation of various frequency components

Represents a signal $f(x)$ in terms of Amplitudes and Phases of its Constituent Sinusoids.

$$f(x) \longrightarrow \boxed{\text{FT}} \longrightarrow F(u)$$

$$f(x) \leftarrow \boxed{\text{IFT}} \leftarrow F(u)$$

Inverse fourier transform (IFT): Image Reconstruction



Computes the **signal** ($f(x)$) from the **Amplitudes** and **Phases** of its Constituent Sinusoids.

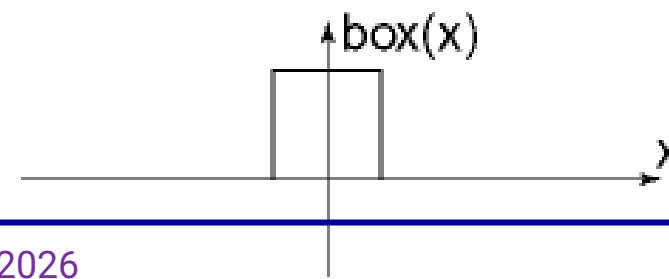
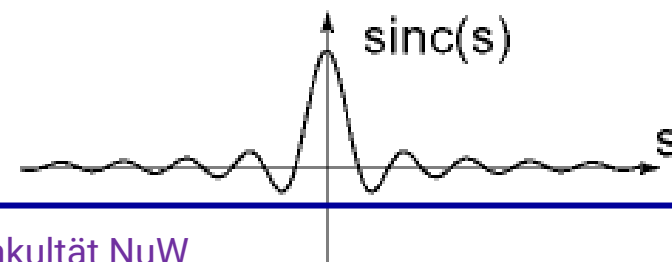
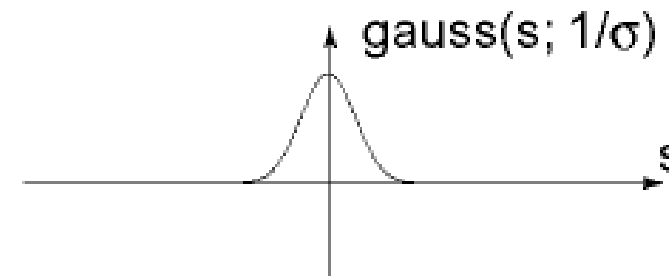
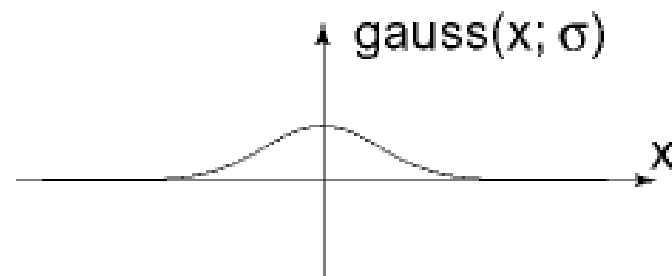
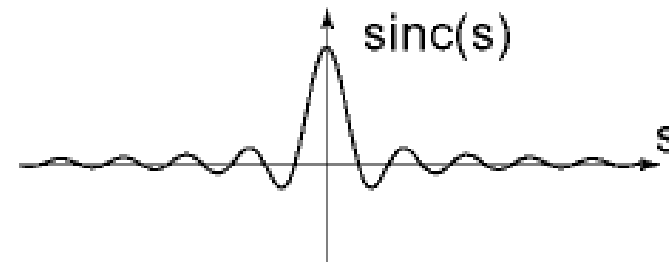
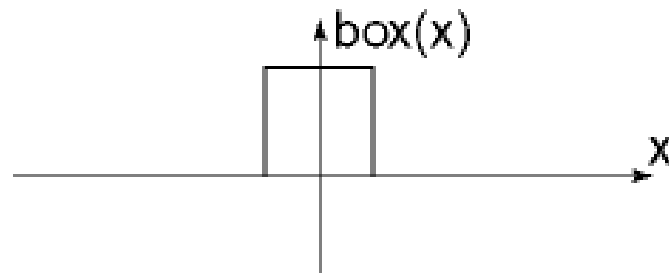
Image Reconstruction:

Both **magnitude** and **phase** are necessary for the **complete reconstruction of the original image** from its frequency domain representation.

Fourier transform pairs

spatial domain

frequency domain

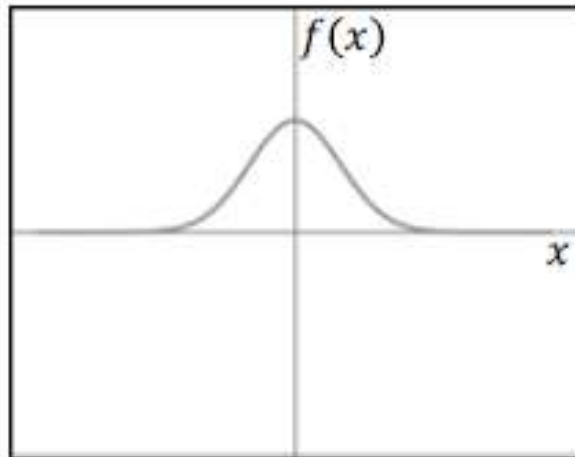


Note :
duality property of
Fourier transform

FT(Gaussian)

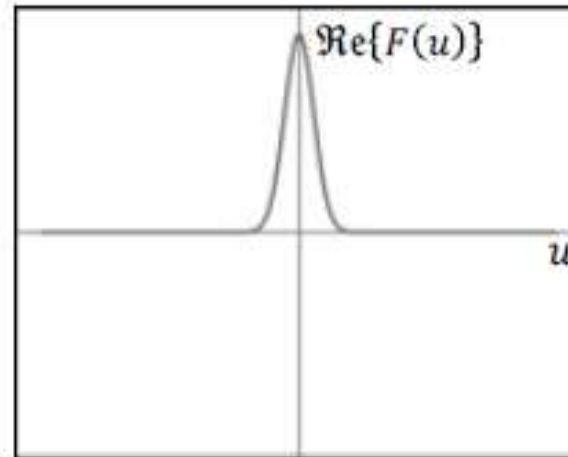


Signal $f(x)$



$$f(x) = e^{-ax^2}$$

Fourier Transform $F(u)$



$$F(u) = \sqrt{\pi/a} e^{-\pi^2 u^2 / a}$$

FT(Gaussian) is also a Gaussian.

The width is proportional to $1/a$.

For any function, if we widen it in the spatial domain, we actually compress it in the frequency domain and vice versa.

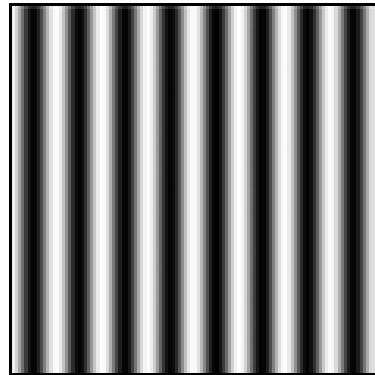
This is referred to as inverse scaling.



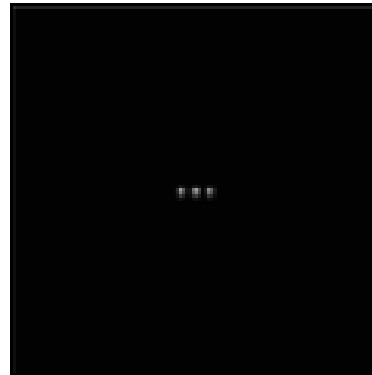
Properties of Fourier Transform

Property	Spatial Domain	Frequency Domain
Linearity	$\alpha f_1(x) + \beta f_2(x)$	$\alpha F_1(u) + \beta F_2(u)$
Scaling	$f(ax)$	$\frac{1}{ a } F\left(\frac{u}{a}\right)$
Shifting	$f(x - a)$	$e^{-i2\pi ua} F(u)$
Differentiation	$\frac{d^n}{dx^n} (f(x))$	$(i2\pi u)^n F(u)$

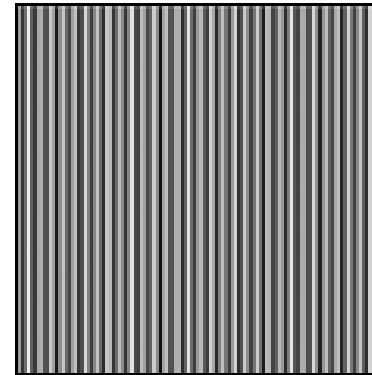
2D Fourier Transform: Example



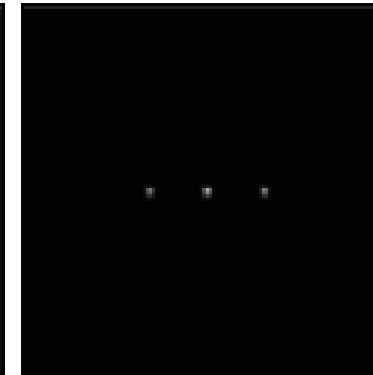
$f(m, n)$



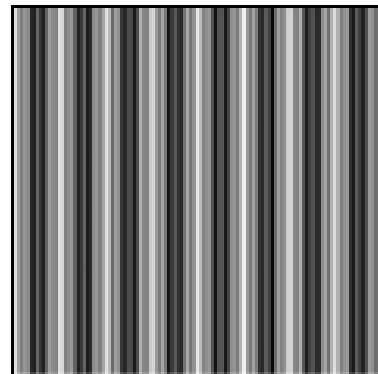
$\log(|F(p, q)|)$



$g(m, n)$



$\log(|G(p, q)|)$



$f(m, n) + g(m, n)$



$\log(|F(p, q) + G(p, q)|)$

compress the wide range
of magnitude values into
a smaller range

Note: $\log()$ is used just for display

How would you generate this Image?:

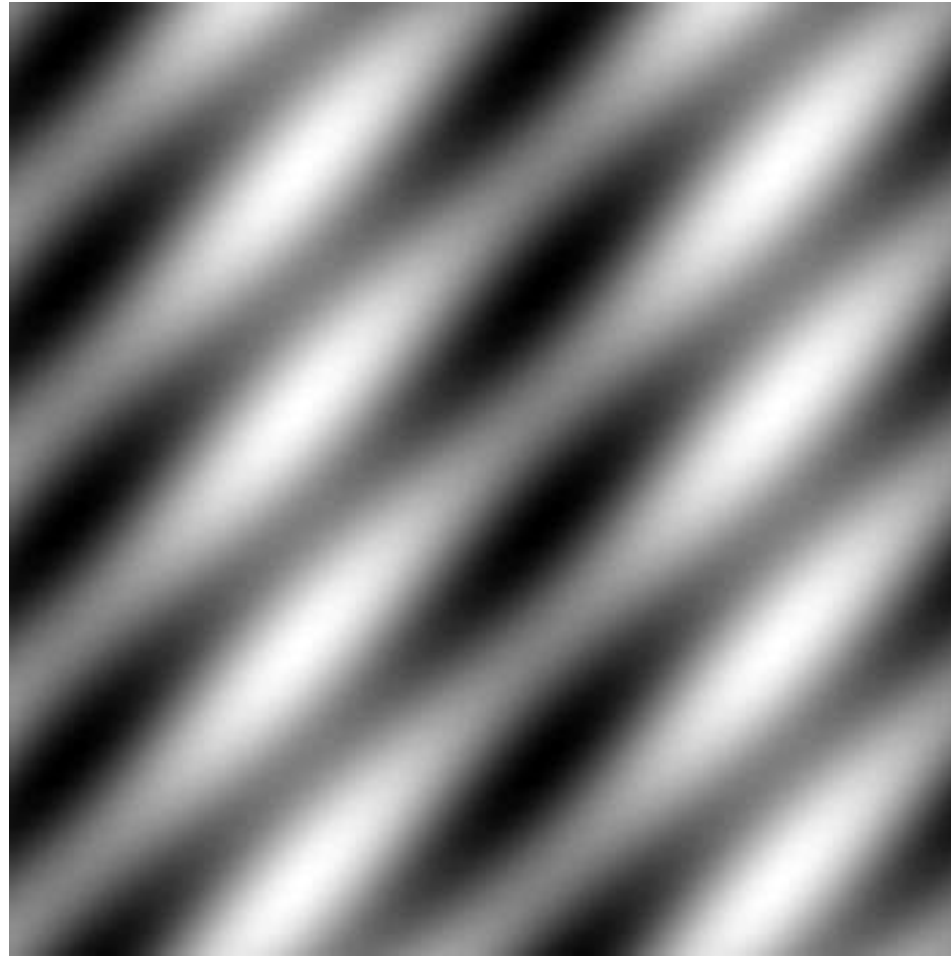
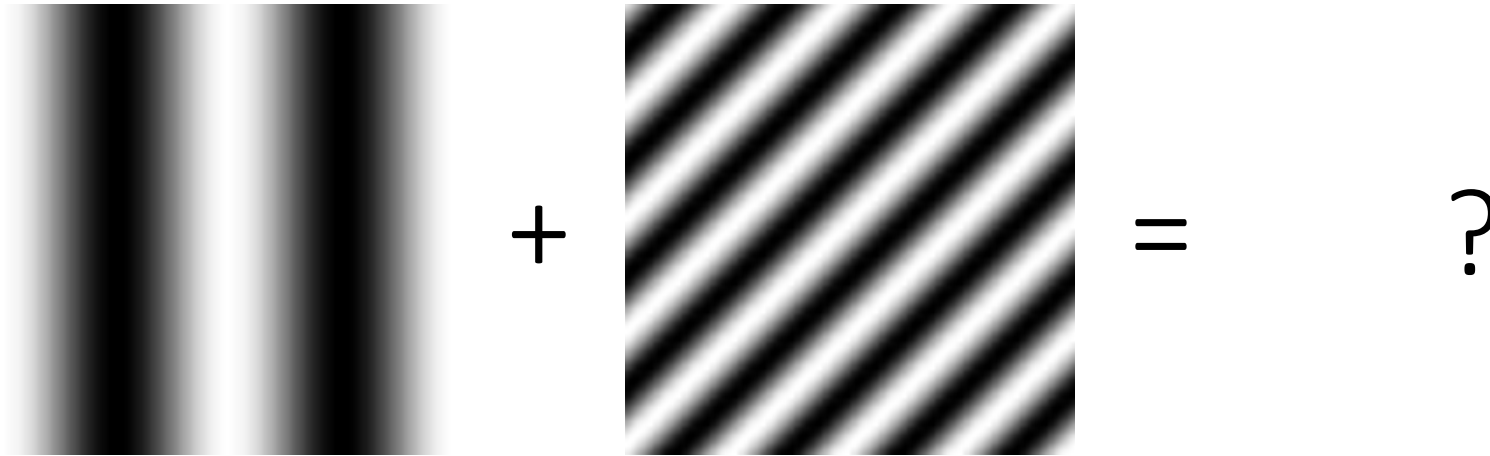
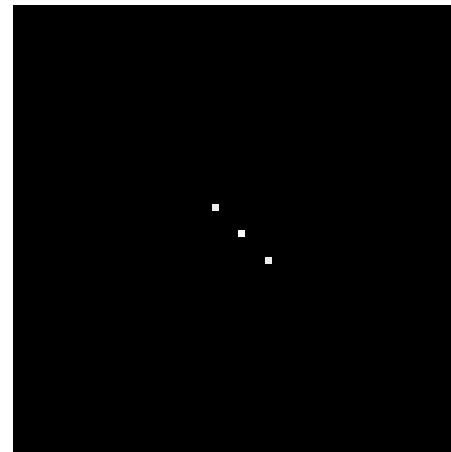
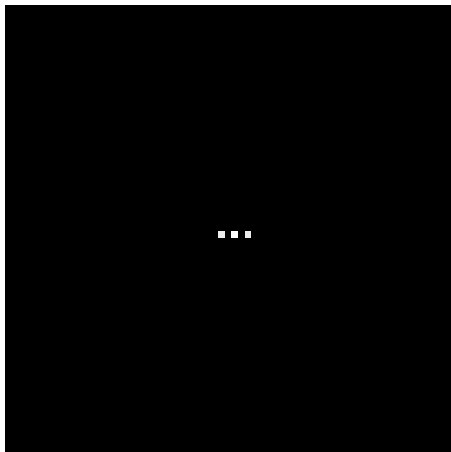
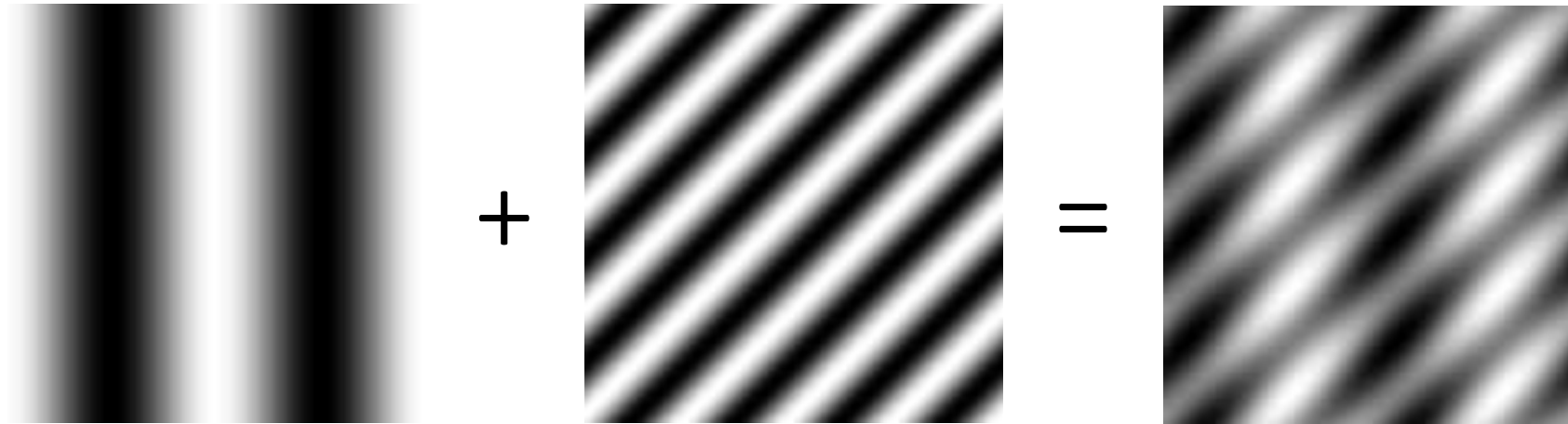


Image Generation and Frequency Domain Visualization:

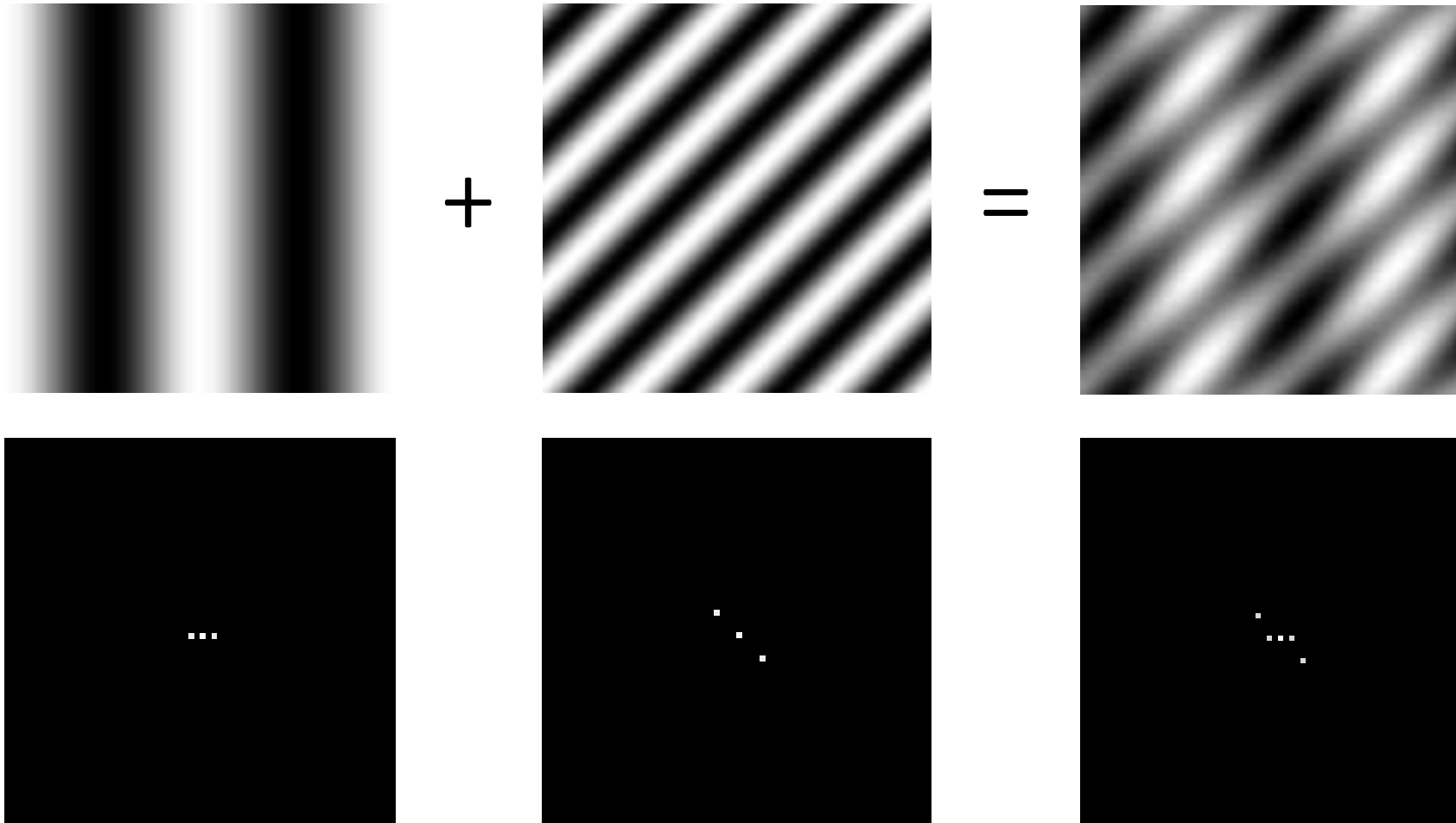


Phase Importance:



?

Frequency Domain Visualization:

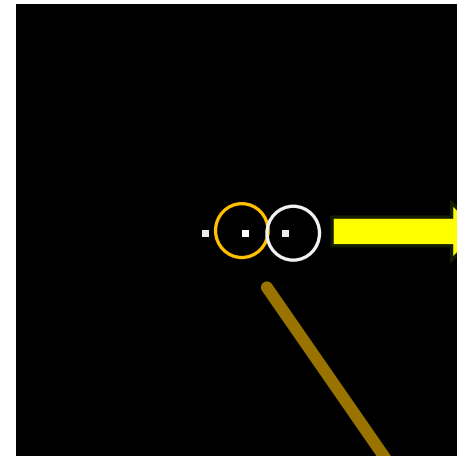


Frequency Domain Visualization:

Spatial domain visualization



Frequency domain visualization



The bright pixels in frequency image (right) *indicate strong intensity at the corresponding positions.*

The high frequency domain, includes those *sharp edges* or *some structure* with the changes of *intensity dramatically along one orientation* (such as the hairs, boundary of hat, etc)



The central part includes the *low frequency components.*

The low frequency domain corresponds to the *smooth areas in the image* (such as skin, hat, background, etc).

Fourier phase importance in reconstruction



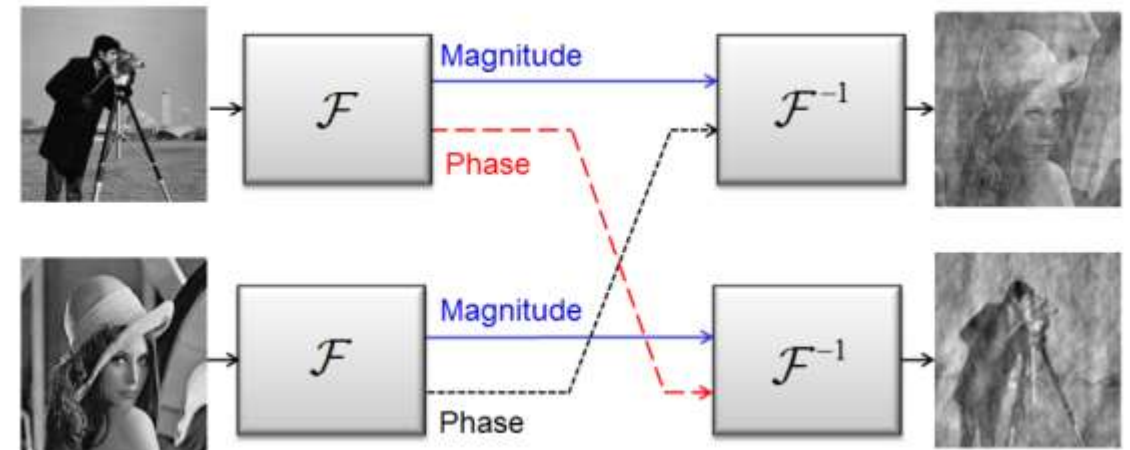
Original Image



Magnitude Preserved,
Phase Set to Zero

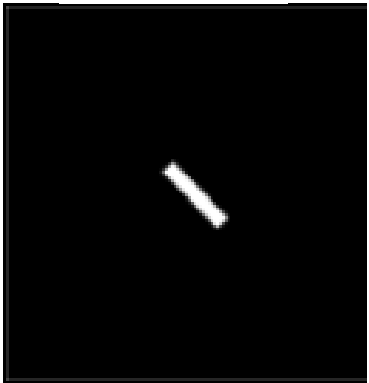


Phase Preserved,
Magnitude Set to Average
of Natural Images

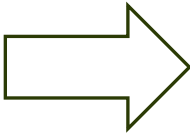


2D-Fourier Transform: Example 1

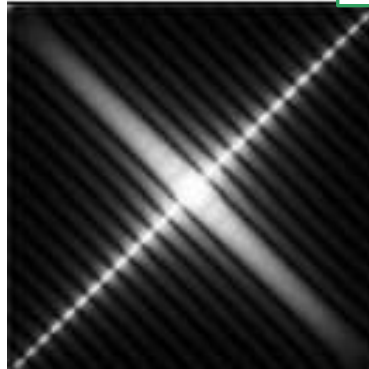
$f(m, n)$



2D-FT



$\log(|F(p, q)|)$

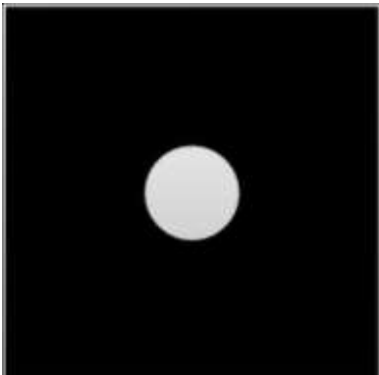


the intensity at each point in the transform image is proportional to the log of the magnitude of the transform value for the corresponding frequency

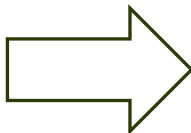


we get very strong frequencies along two lines.

we have two pairs of strong edges and to re-create each edge we need frequencies in the direction perpendicular to the edge



2D-FT



The flat part of the disk produces some low frequencies.

The edges produce high frequencies.

Note

image is rotationally symmetric, its Fourier transform is rotationally symmetric as well.

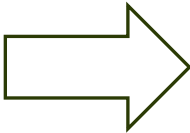
Min  Max

2D-Fourier Transform: Example 2

$f(m, n)$



2D-FT



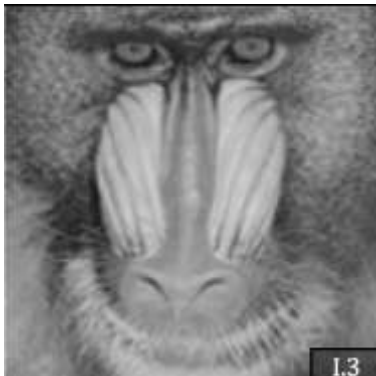
$\log(|F(p, q)|)$



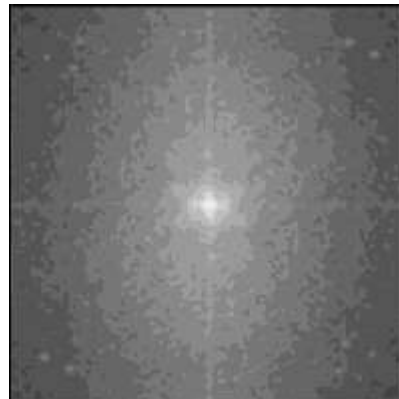
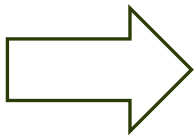
the intensity at each point in the transform image is proportional to the log of the magnitude of the transform value for the corresponding frequency

we see three strong lines in the Fourier transform, indicating high frequencies along these lines.

This is because the original image has edges along three dominant directions: vertical edges, horizontal edges, and slanted edges



2D-FT



Mandrill, there is not really much that can be said.

The image is complex it is hard to interpret its transform.

As humans, we are not wired to easily interpret signals in frequency domain

Min  Max

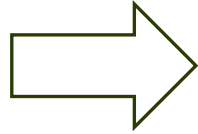
2D-Fourier Transform: Example 3



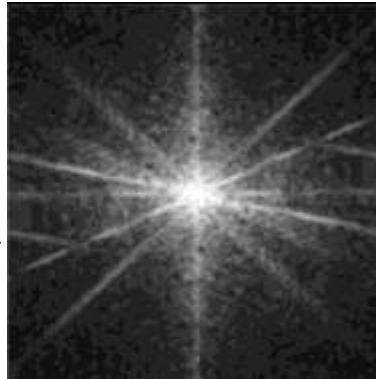
$f(m, n)$



2D-FT



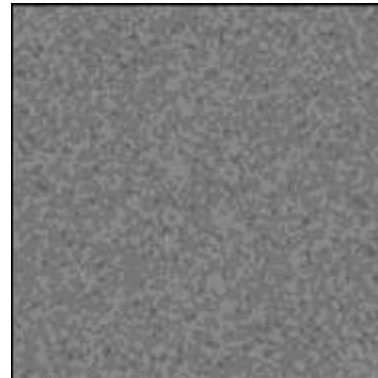
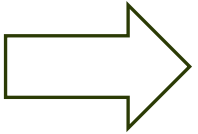
$\log(|F(p, q)|)$



more complex image with lots of strong edges



2D-FT



The image which is just noise.

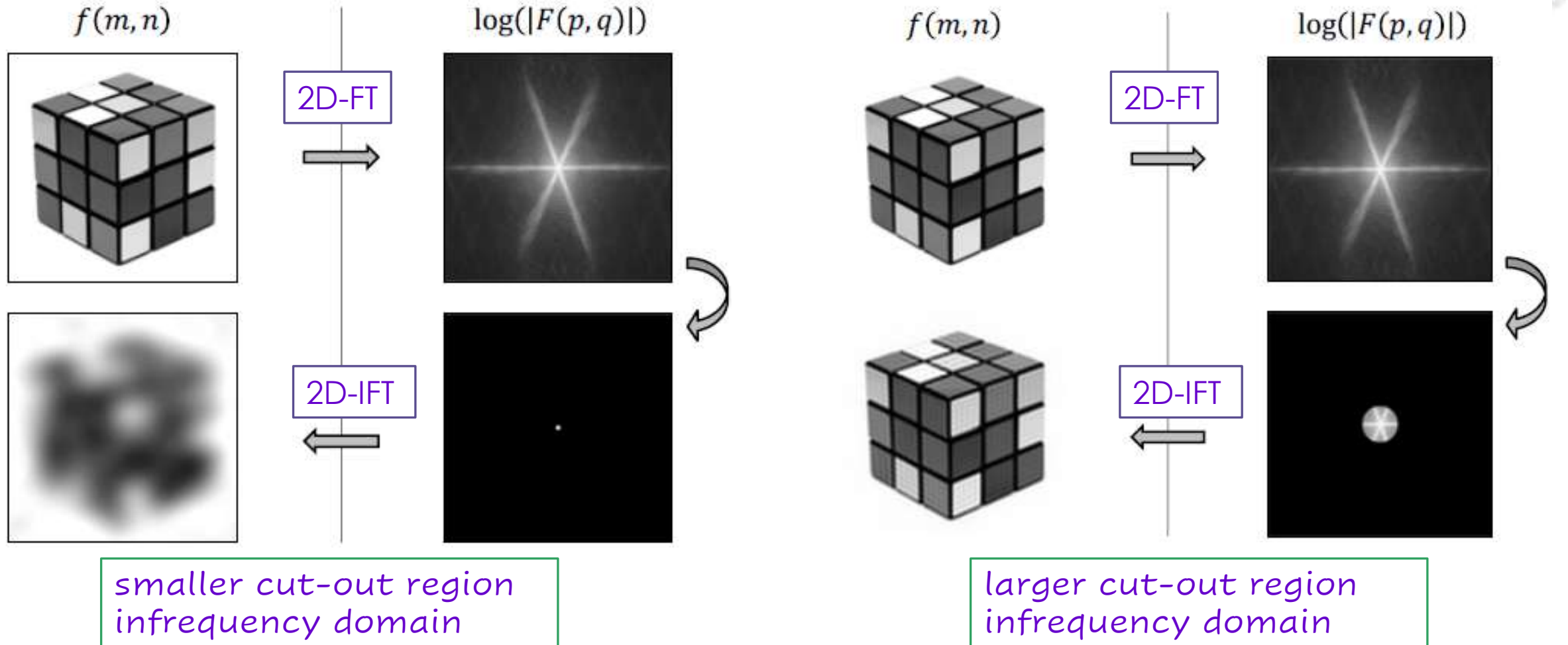
In this case we see that we get very strong values in the Fourier transform for high frequencies.

Min  Max

we want to low-pass filter this image, we can simply cut out the outer parts of the Fourier transform.

Low Pass Filtering

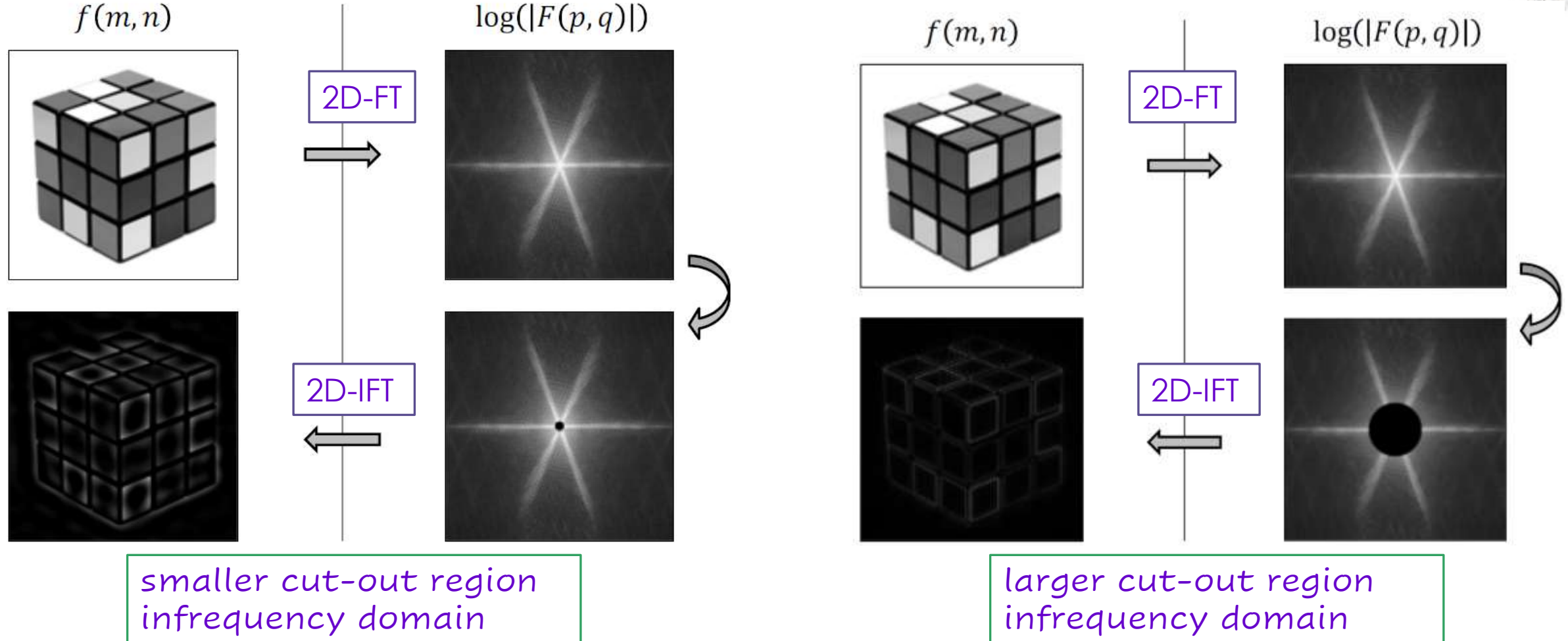
Taking the inverse Fourier transform, we see that we get a blurrier (smoother) version of the image



high-pass filter the image. We eliminate the low frequencies and keep the high frequencies.

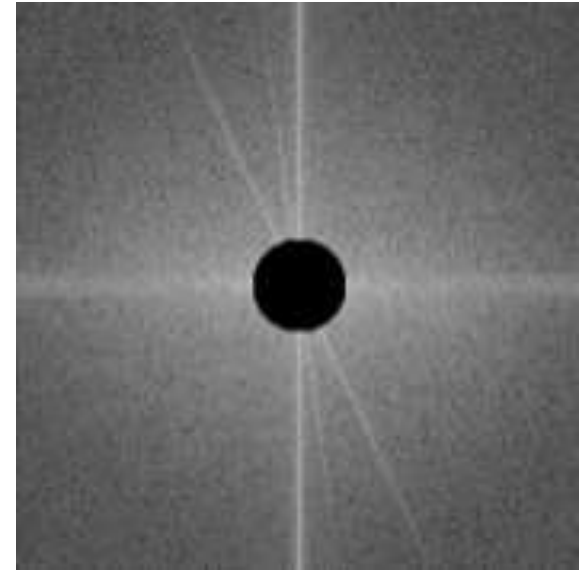
High Pass Filtering

Remove all of the constant brightness regions in the image, leaving us with the regions where there is a rapid change in brightness



More filtering examples

high-pass

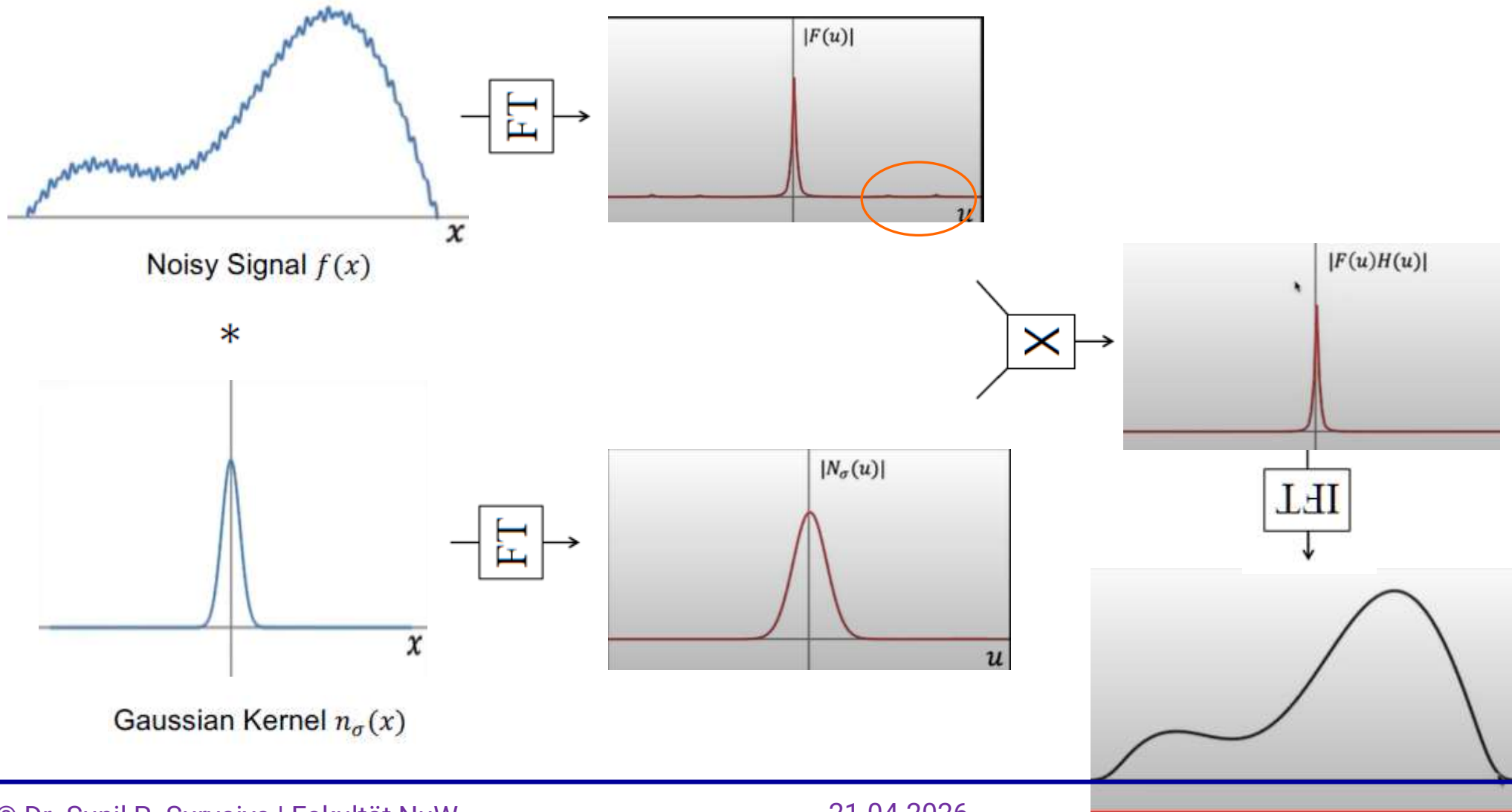


Convolution and Fourier Transform:



$$\begin{array}{ccccc}
 g(x) & = & f(x) & * & h(x) \\
 \uparrow & & | & & | \\
 \boxed{\text{IFT}} & & \boxed{\text{FT}} & & \boxed{\text{FT}} \\
 | & & \downarrow & & \downarrow \\
 G(u) & = & F(u) & \times & H(u)
 \end{array}$$

Gaussian smooting:



Gaussian Smoothing:

